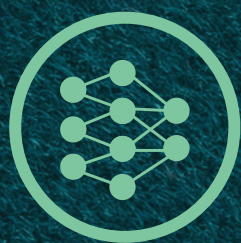


MNIST IMAGE RECOGNITION WITH CNNs USING ->



TENSORFLOW
KERAS



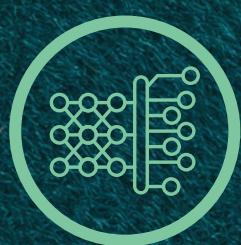
THE OBJECTIVE

Develop an efficient, lightweight neural network to identify handwritten digits (0-9) using MNIST dataset.



MNIST DATASET

70,000 grayscale images of handwritten digits (0-9) sized 28x28 pixels. Widely known, used and tested for training and evaluating machine learning models in image classification.



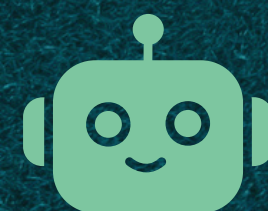
CONVNET ARCHITECTURE

This model uses a Convolutional Neural Network, which are specialised for Image classification by finding patterns in images with **Kernels** and representing these patterns with **Feature Maps**.

HANDWRITING RECOGNITION

3

ROBOTICS AUTOMATIONS



WEB APPLICATIONS



COMPUTER VISION

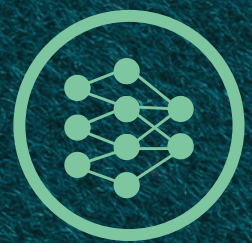


SELF-DRIVING CARS



TARGETED APPLICATIONS

Architecture OF CONVOLUTIONAL NEURAL NETWORK



INPUT LAYER

Inputting Images from MNIST dataset with a 3-dimensional tensor represented by a Numpy Array (28x28x1)



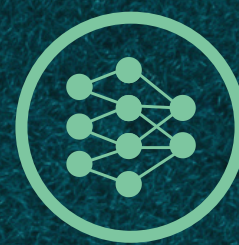
1ST COV LAYER

128 Kernels/Filters that are 3x3 to pickup trends in the image, activated with a ReLU function



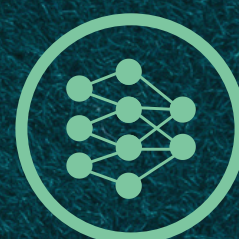
2ND COV LAYER

256 Kernels/Filters that are 3x3 to pickup trends in the image, activated with a ReLU function



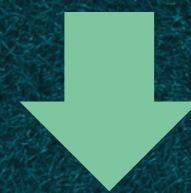
FLATTEN

Converting 2D feature maps into 1D Feedforward Neural Network



DROPOUT LAYER

A Dropout layer (50%) to prevent overfitting.



OUTPUT LAYER

Dense output layer with 10 neurons to output prediction of each class, with probability sorted with a SoftMax function

Testing Results



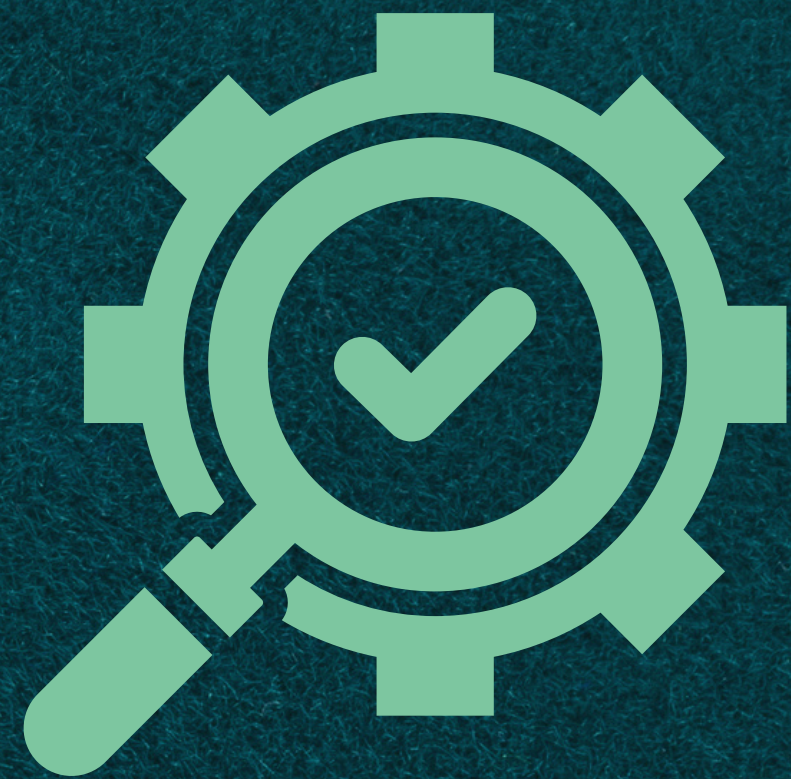
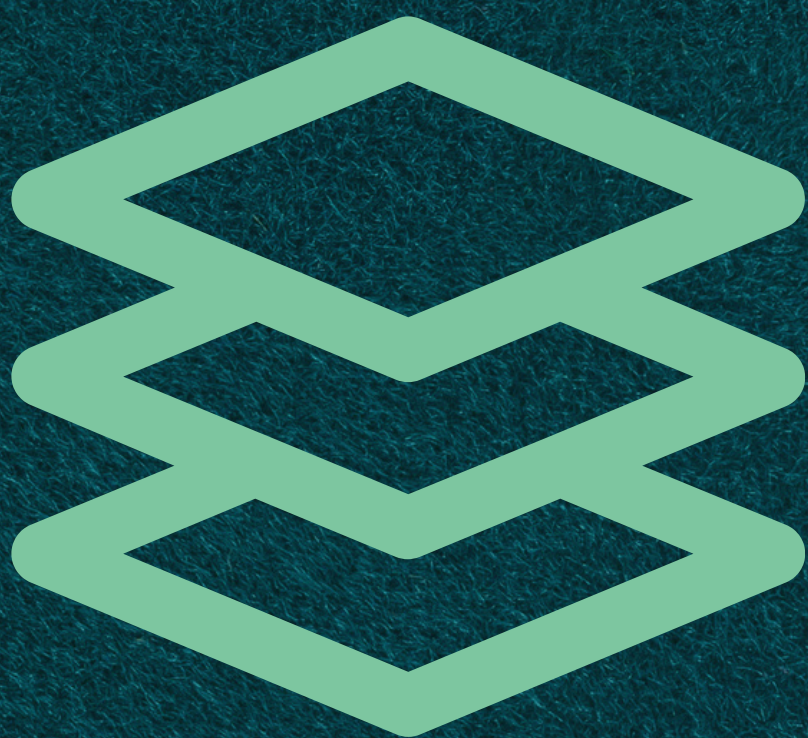
ITERATION 1

VANILLA VERSION OF THE NETWORK

Data Batch Size: 8
Image Data: float32, repeated
Epochs: 10
Hardware: NVIDIA A100

Overall Accuracy over 10,000 test samples:

Loss Function - 0.14181962609291077
Accuracy - 99.21000003814697%



ITERATION 2

Optimised Network with image data type as **float64** rather than 32, theoritcally enabling bettery accuracy within the network.

Data Batch Size: 8
Image Data: float64, repeated
Epochs: 10
Hardware: NVIDIA A100

Overall Accuracy over 10,000 test samples (UNCHANGED):

Loss Function - 0.14181962609291077
Accuracy - 99.21000003814697%

THUS, INCREASING MODEL DATA TYPE ACCURACY DID NOT EFFECTIVELY INFLUENCE PERFORMANCE